# Approximation Algorithms

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# Outline


1. Overview
2. Vertex-cover problem
3. TSP
4. Set-covering problem
5. Randomization and LP
6. Subset-sum problem

# Performance ratios for approximation algorithms

## Definition

We say that an algorithm for a problem has an ***approximation ratio*** of $\rho(n)$ if, for any input of size $n$, the cost $C$ of the solution produced by the algorithm is within a factor of $\rho(n)$ of the cost $C^*$ of an optimal solution:

$$\max\left(\frac{C}{C^*}, \frac{C^*}{C}\right) \leq \rho(n).$$

# Performance ratios for approximation algorithms

## Definition

- An ***approximation scheme*** for an optimization problem is an approximation algorithm that takes as input not only an instance of the problem, but also a value $\epsilon > 0$ such that for any fixed $\epsilon$, the scheme is a $(1 + \epsilon)$-approximation algorithm.

# Performance ratios for approximation algorithms

## Definition

- We say that an approximation scheme is a ***polynomial-time approximation scheme*** if for any fixed $\epsilon > 0$, the scheme runs in time polynomial in the size $n$ of its input instance.

# Performance ratios for approximation algorithms

## Definition

- We say that an approximation scheme is a ***fully polynomial-time approximation scheme*** if it is an approximation scheme and its running time is polynomial both in $1/\epsilon$ and in the size $n$ of the input instance.

  For example, the scheme might have a running time of $O((1/\epsilon)^2 n^3)$.

# The vertex-cover problem

## Definition

- A ***vertex cover*** of an undirected graph $G = (V, E)$ is a subset $V' \subseteq V$ such that if $(u, v)$ is an edge of $G$, then either $u \in V'$ or $v \in V'$ (or both).
- The ***vertex-cover problem*** is to find a vertex cover of minimum size in a given undirected graph. We call such a vertex cover an ***optimal vertex cover***.

# The vertex-cover problem

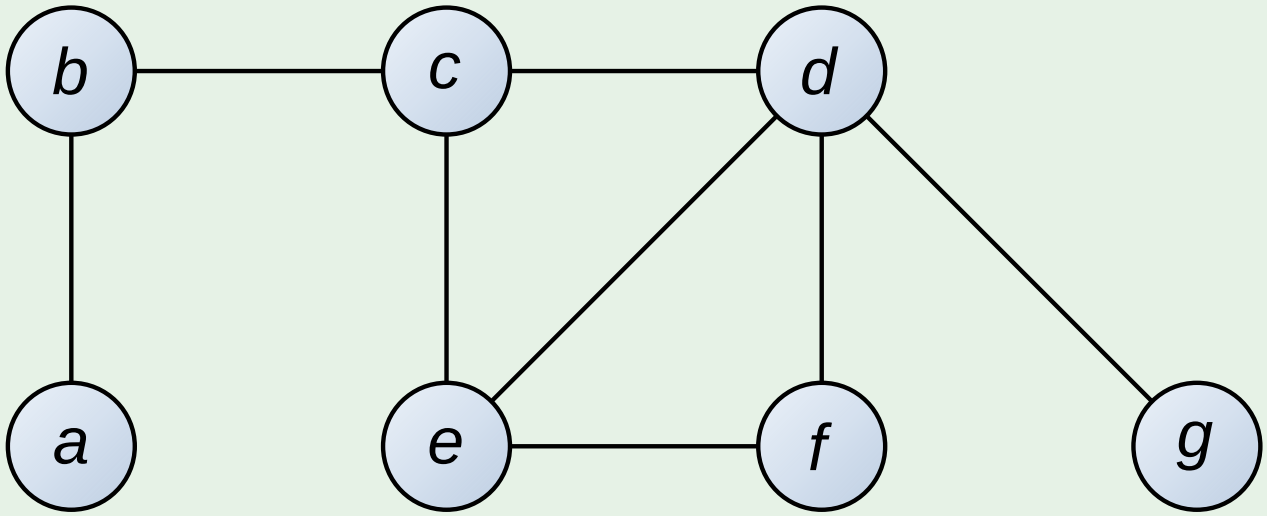
APPROX-VERTEX-COVER($G$)

```
C ← ∅
E′ ← E[G]
while E′ ≠ ∅
        do let (u, v) be an arbitrary edge of E′
            C ← C ∪ {u, v}
            remove from E′ every edge incident
            on either u or v
return C
```

Skip Example

# The vertex-cover problem

## Example



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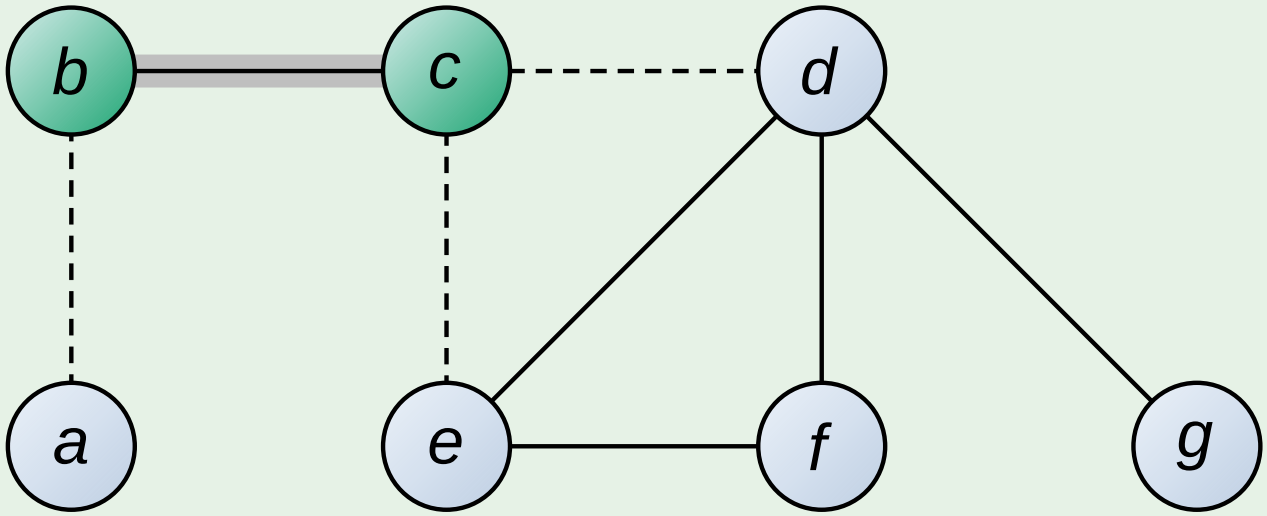
## Example



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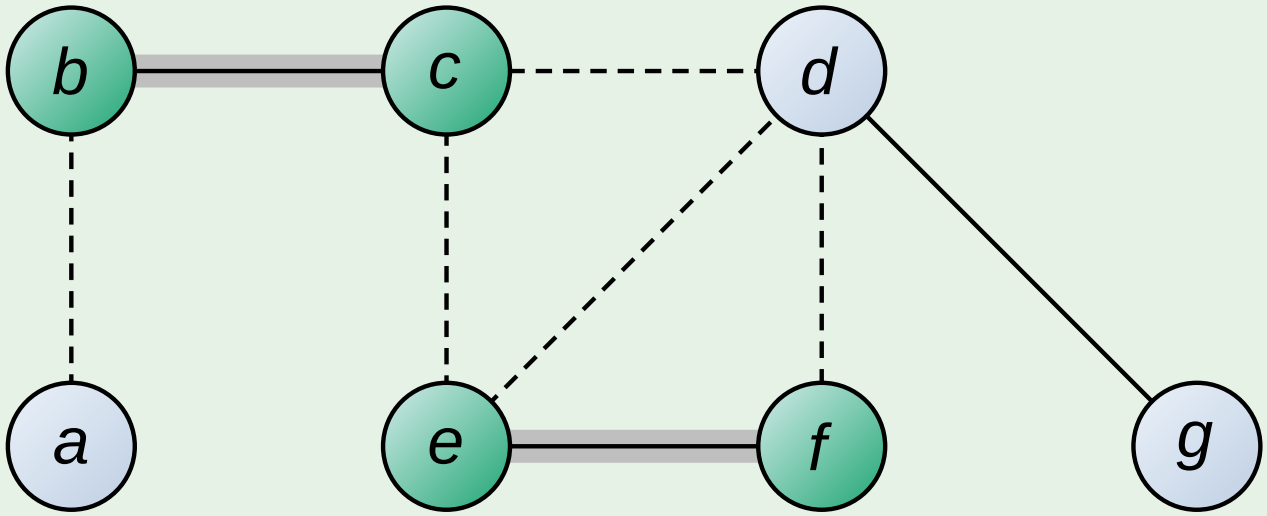

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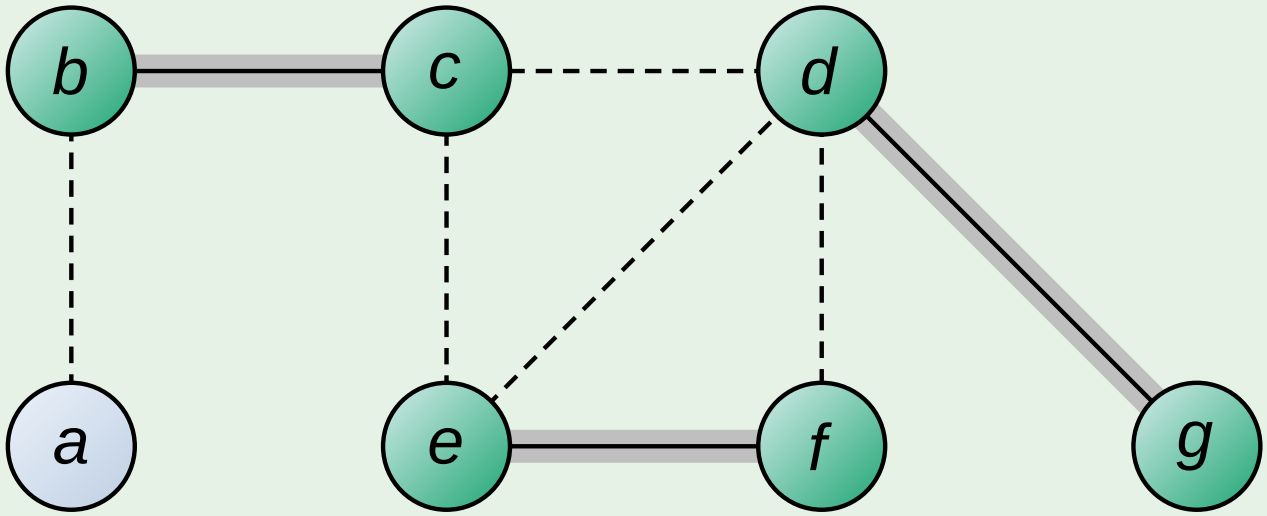

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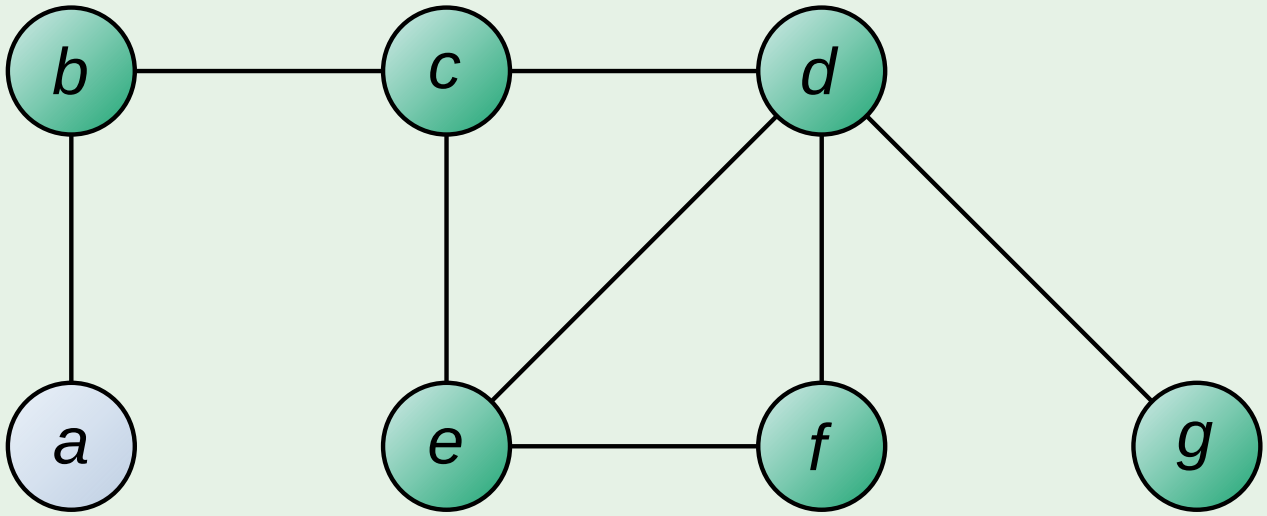

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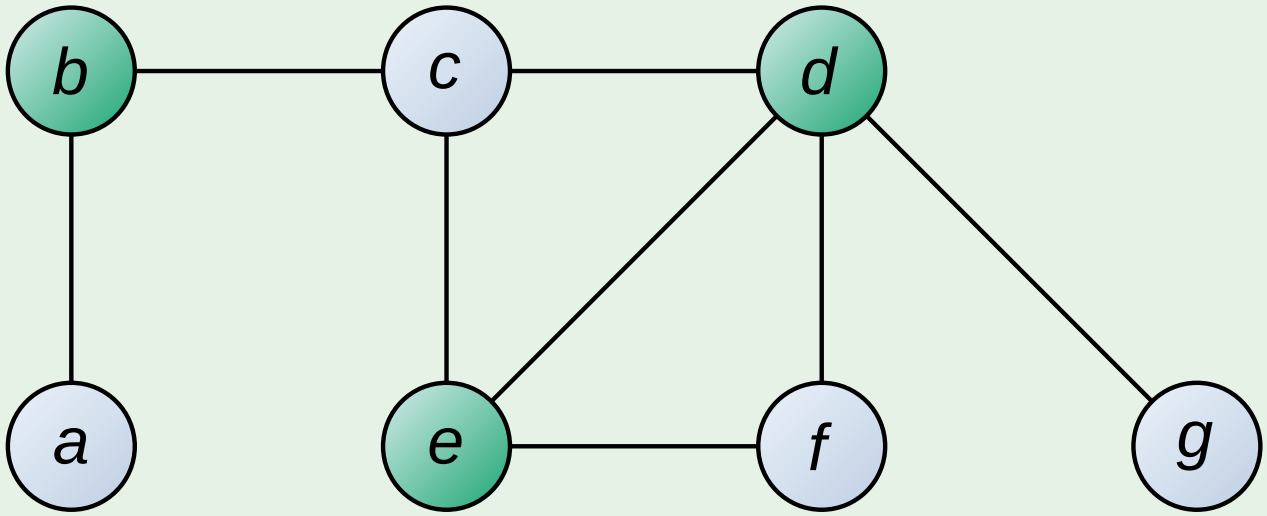

# The vertex-cover problem

**Theorem 35.1**

APPROX-VERTEX-COVER is a polynomial-time 2-approximation algorithm.

**Proof.**

- The running time of this algorithm is $O(V + E)$, using adjacency lists to represent $E'$.

# The vertex-cover problem

**Theorem 35.1**

APPROX-VERTEX-COVER is a polynomial-time 2-approximation algorithm.

**Proof.**

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# The vertex-cover problem

**Proof.**

- Let $A$ denote the set of edges that were picked in line 4 of APPROX-VERTEX-COVER. Thus we have $|C^*| \geq |A|$.
- $|C| = 2|A| \leq 2|C^*|$

□

# The traveling-salesman problem

## Definition

$$\textbf{TSP} = \{\langle G, c, k\rangle : G = (V, E) \text{ is a complete graph, } c \text{ is a function from } V \times V \to \mathbb{Z}, k \in \mathbb{Z}, \text{ and } G \text{ has a traveling-salesman tour with cost at most } k\}$$

# TSP with triangle inequality

## Triangle inequality

- Let $c(A)$ denote the total cost of the edges in the subset $A \subseteq E$:

$$c(A) = \sum_{(u,v) \in A} c(u,v).$$

# TSP with triangle inequality

## Triangle inequality

- We say that cost function $c$ satisfies the ***triangle inequality*** if for all vertices $u, v, w \in V$,

$$c(u, w) \leq c(u, v) + c(v, w).$$

# TSP with triangle inequality

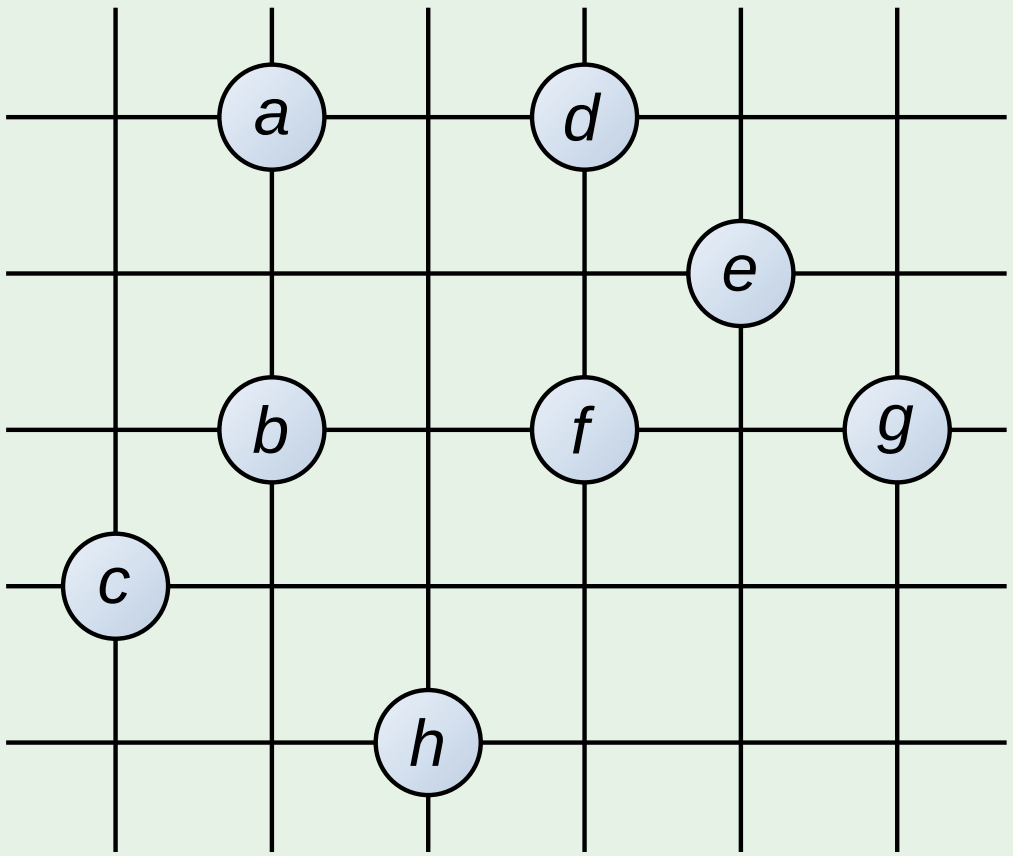
APPROX-TSP-TOUR$(G, c)$

1 select a vertex $r \in V[G]$ to be a "root" vertex
2 compute a minimum spanning tree $T$ for $G$
  from root $r$ using MST-PRIM$(G, c, r)$
3 let $L$ be the list of vertices visited in
  a preorder tree walk of $T$
4 **return** the hamiltonian cycle $H$ that
  visits the vertices in the order $L$

Skip Example

# TSP with triangle inequality

## Example



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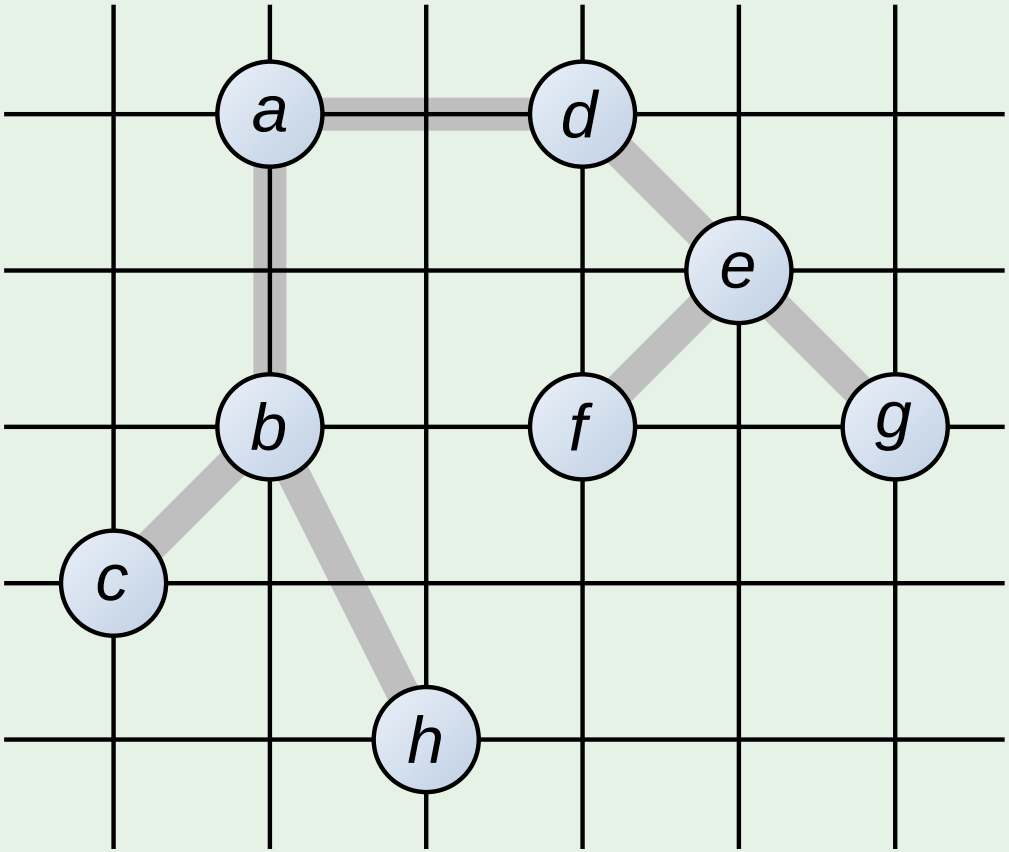

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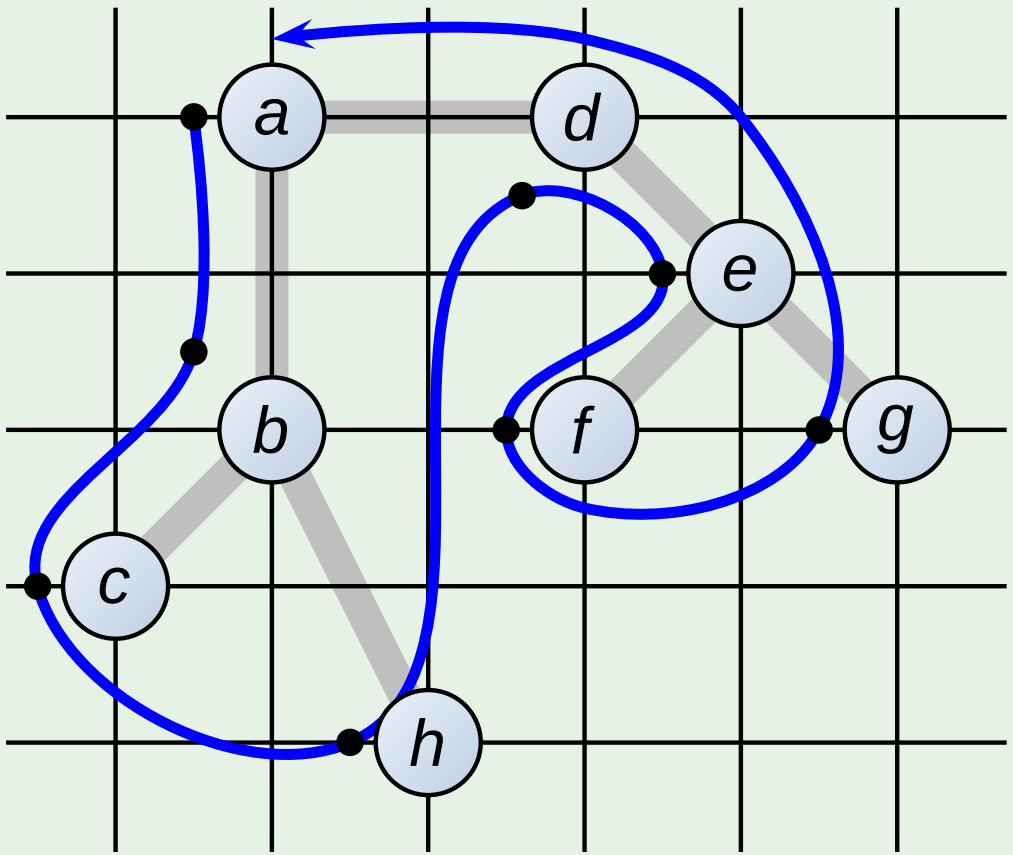

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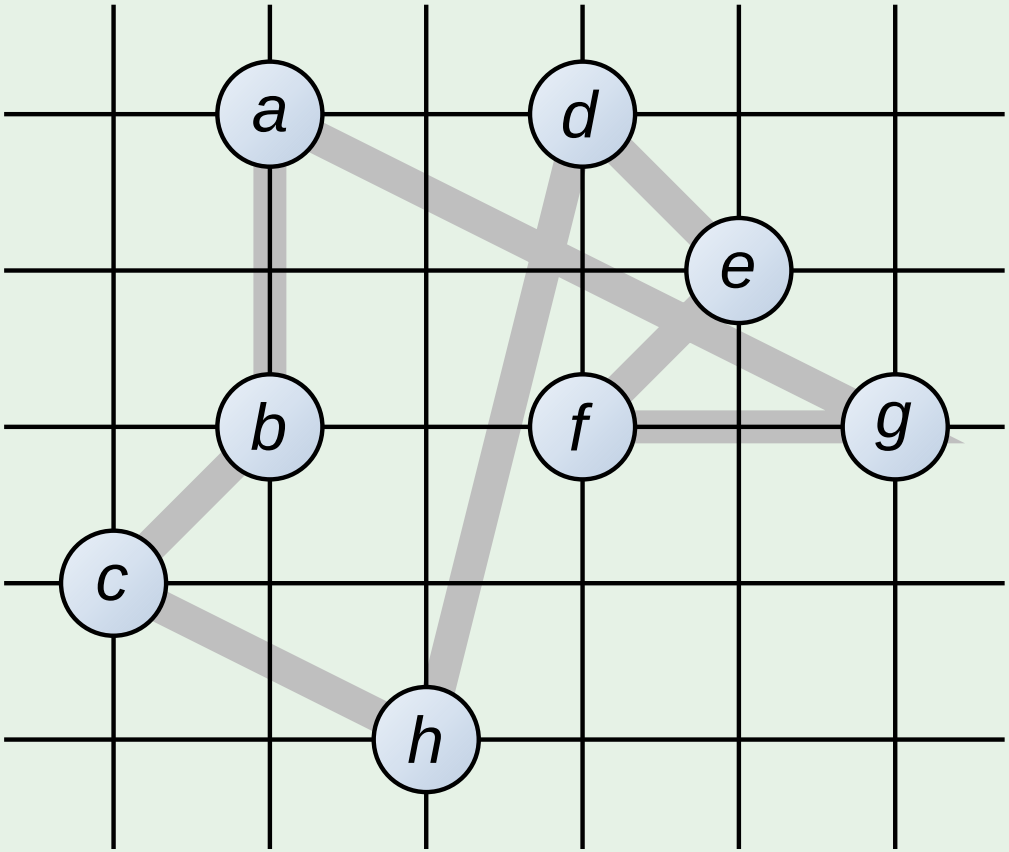

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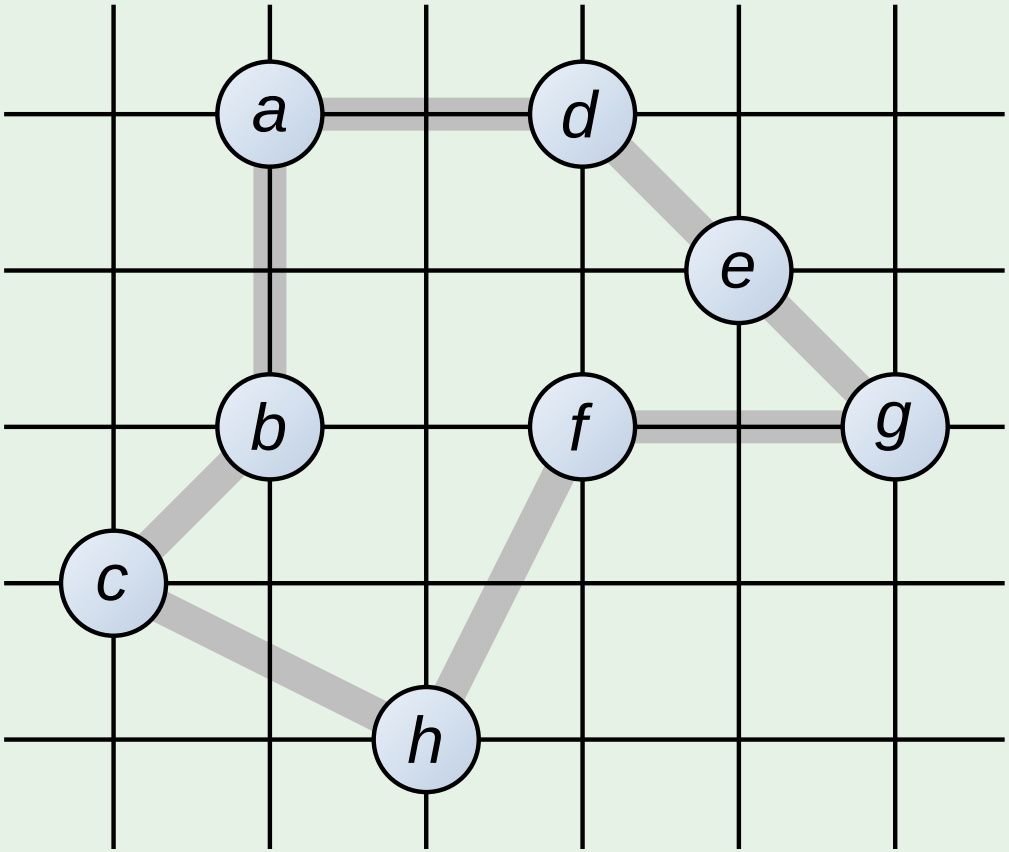

# TSP with triangle inequality

**Theorem 35.2**

APPROX-TSP-TOUR is a polynomial-time 2-approximation algorithm for the traveling-salesman problem with the triangle inequality.

# TSP with triangle inequality

**Proof.**

- The running time of APPROX-TSP-TOUR is $\Theta(V^2)$.
- $c(T) \leq c(H^*)$
- A ***full walk*** of $T$ lists the vertices when they are first visited and also whenever they are returned to after a visit to a subtree. Let us call this walk $W$ .

$$c(W) = 2c(T) \leq 2c(H^*).$$

□

# TSP with triangle inequality

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# TSP with triangle inequality

**Proof.**

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- $c(T) \leq c(H^*)$
- A ***full walk*** of $T$ lists the vertices when they are first visited and also whenever they are returned to after a visit to a subtree. Let us call this walk $W$ .
  $c(H) \leq c(W) = 2c(T) \leq 2c(H^*)$.

$\square$

# The general TSP

**Theorem 35.3**

If $P \neq NP$, then for any constant $\rho \geq 1$, there is ***no*** polynomial-time approximation algorithm with approximation ratio $\rho$ for the general traveling-salesman problem.

# The general TSP

**Proof.**

- Suppose to the contrary that for some number $\rho \geq 1$, there is a polynomial-time approximation algorithm $A$ with approximation ratio $\rho$. Without loss of generality, we assume that $\rho$ is an integer, by rounding it up if necessary.
- Let $G = (V, E)$ be an instance of the hamiltonian-cycle problem.

# The general TSP

**Proof.**

- Let $G' = (V, E')$ be the complete graph on $V$ ; that is, $E' = \{(u, v) : u, v \in V \text{ and } u \neq v\}$.
- Assign an integer cost to each edge in $E'$ as follows:

$$c(u, v) = \begin{cases} 1 & \text{if } (u, v) \in E \\ \rho|V| + 1 & \text{otherwise.} \end{cases}$$

- $(\rho|V| + 1) + (|V| - 1) = \rho|V| + |V| > \rho|V|$

$\square$

# The general TSP

**Proof.**

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- $(\rho|V| + 1) + (|V| - 1) = \rho|V| + |V| > \rho|V|$

$\square$

# The set-covering problem

## Definition

- An instance $(X, \mathcal{F})$ of the ***set-covering problem*** consists of a finite set $X$ and a family $\mathcal{F}$ of subsets of $X$, such that every element of $X$ belongs to at least one subset in :

$$X = \bigcup_{S \in \mathcal{F}} S.$$

# The set-covering problem

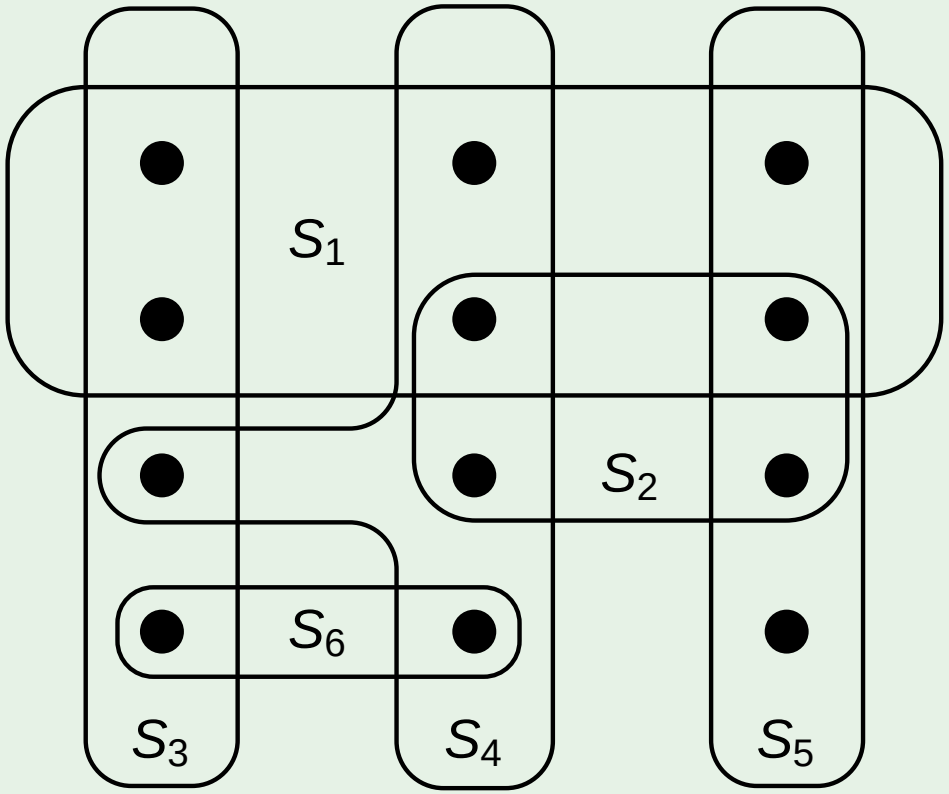
## Definition

- The problem is to find a ***minimum-size*** subset $\mathcal{C} \subseteq \mathcal{F}$ whose members cover all of $X$:

$$X = \bigcup_{S \in \mathcal{C}} S.$$

# The set-covering problem

## Example



# A greedy approximation algorithm

GREEDY-SET-COVER$(X, \mathcal{F})$

```
U ← X
C ← ∅
while U ≠ ∅
      do select an S ∈ F
            that maximizes |S ∩ U|
            U ← U − S
            C ← C ∪ {S}
return C
```

# A greedy approximation algorithm

**Theorem 35.4**

GREEDY-SET-COVER is a polynomial-time $\rho(n)$-approximation algorithm, where $\rho(n) = H(\max\{|S| : S \in \mathcal{F}\})$.
We define $H(d) = \sum_{i=1}^{d} 1/i$ and $H(0) = 0$.

# A greedy approximation algorithm

**Proof.**

- The algorithm GREEDY-SET-COVER can easily be implemented to run in time polynomial in $|X|$ and $|\mathcal{F}|$. For example, there is an implementation that runs in time $O(|X||\mathcal{F}|\min(|X|,|\mathcal{F}|))$.

# A greedy approximation algorithm

**Proof.**

- Let $S_i$ denote the $i$th subset selected by GREEDY-SET-COVER; the algorithm incurs a cost of $1$ when it adds $S_i$ to $\mathcal{C}$. We spread this cost of selecting $S_i$ evenly among the elements covered for the first time by $S_i$.

# A greedy approximation algorithm

**Proof.**

- Let $c_x$ denote the cost allocated to element $x$, for each $x \in X$. If $x$ is covered for the first time by $S_i$, then

$$c_x = \frac{1}{|S_i - (S_1 \cup S_2 \cup \cdots \cup S_{i-1})|}$$

# A greedy approximation algorithm

**Proof.**

- At each step of the algorithm, 1 unit of cost is assigned, and so $|\mathcal{C}| = \sum_{x \in X} c_x$.
- The cost assigned to the optimal cover is $\sum_{S \in \mathcal{C}^*} \sum_{x \in S} c_x$,

# A greedy approximation algorithm

**Proof.**

- and since each $x \in X$ is in at least once set $S \in \mathcal{C}^*$, we have $|\mathcal{C}| = \sum_{x \in X} c_x \leq \sum_{S \in \mathcal{C}^*} \sum_{x \in S} c_x$.
- The remainder of the proof rests on the following key inequality, which we shall prove shortly. For any set $S$ belonging to the family , $\sum_{x \in S} c_x \leq H(|S|)$.
- $|\mathcal{C}| \leq \sum_{S \in \mathcal{C}^*} H(|S|) \leq |\mathcal{C}^*| \cdot H(\max |S|)$.

# A greedy approximation algorithm

**Proof.**

- Consider any set $S \in \mathcal{F}$ and $i = 1, 2, \ldots, |\mathcal{C}|$, and let $u_i = |S - (S_1 \cup S_2 \cup \ldots \cup S_i)|$ be the number of elements in $S$ remaining uncovered after $S_1, S_2, \ldots, S_i$ have been selected by the algorithm.

# A greedy approximation algorithm

**Proof.**

- We define $u_0 = |S|$ to be the number of elements of $S$, which are all initially uncovered. Let $k$ be the least index such that $u_k = 0$, so that each element in $S$ is covered by at least one of the sets $S_1, S_2, \ldots, S_k$.
- Then, $u_{i-1} \geq u_i$, and $u_{i-1} - u_i$ elements of $S$ are covered for the first time by $S_i$.

# A greedy approximation algorithm

**Proof.**

- Thus

$$\sum_{x \in S} c_x = \sum_{i=1}^{k} (u_{i-1} - u_i) \cdot \frac{1}{|S_i - (S_1 \cup S_2 \cup \cdots \cup S_{i-1})|}$$

# A greedy approximation algorithm

**Proof.**

- Observe that

$$\begin{aligned} & |S_i - (S_1 \cup S_2 \cup \cdots \cup S_{i-1})| \\ \geq\ & |S - (S_1 \cup S_2 \cup \cdots \cup S_{i-1})| \\ =\ & u_{i-1}, \end{aligned}$$

because the greedy choice of $S_i$ guarantees that $S$ cannot cover more new elements that $S_i$ does.

# A greedy approximation algorithm

## Proof.

- Consequently, we obtain

$$\sum_{x \in S} c_x \leq \sum_{i=1}^{k} (u_{i-1} - u_i) \cdot \frac{1}{u_{i-1}} = \sum_{i=1}^{k} \sum_{j=u_i+1}^{u_{i-1}} \frac{1}{u_{i-1}}$$

$$\leq \sum_{i=1}^{k} \sum_{j=u_i+1}^{u_{i-1}} \frac{1}{j} \qquad (\textit{because } j \leq u_{i-1})$$

$$= \sum_{i=1}^{k} \left( \sum_{j=1}^{u_{i-1}} \frac{1}{j} - \sum_{j=1}^{u_i} \frac{1}{j} \right)$$

# A greedy approximation algorithm

**Proof.**

- Thus,

$$\begin{aligned}\sum_{x \in S} c_x &\leq \sum_{i=1}^{k} (H(u_{i-1}) - H(u_i)) \\ &= H(u_0) - H(u_k) \\ &= H(u_0) \\ &= H(|S|).\end{aligned}$$

□

# A greedy approximation algorithm

**Corollary 35.5**

GREEDY-SET-COVER is a polynomial-time $(\ln|X| + 1)$-approximation algorithm

# Randomization

## Definition

We say that an algorithm for a randomized problem has an ***approximation ratio*** of $\rho(n)$ if, for any input of size $n$, the ***expected*** cost $C$ of the solution produced by the algorithm is within a factor of $\rho(n)$ of the cost $C^*$ of an optimal solution:

$$\max\left(\frac{C}{C^*}, \frac{C^*}{C}\right) \leq \rho(n).$$

# Randomization

## Definition

We also call a randomized algorithm that achieves an approximation ratio of $\rho(n)$ a $\rho(\mathbf{n})$ ***randomized approximation algorithm***.

# Randomization

**Theorem 35.6**

Given an instance of ***MAX-3-CNF*** satisfiability with $n$ variables $x_1, x_2, \ldots, x_n$ and $m$ clauses, the ***randomized algorithm*** that independently sets each variable to $1$ with probability $1/2$ and to $0$ with probability $1/2$ is a randomized $8/7$-approximation algorithm.

# Randomization

**Proof.**

For $i = 1, 2, \ldots, m$, we define the indicator random variable

$$Y_i = I\{ \text{ clause } i \text{ is satisfied } \}$$

Let $Y$ be the number of satisfied clauses overall, so that $Y = Y_1 + Y_2 + \cdots + Y_m$.

# Randomization

**Proof.**

For $i = 1, 2, \ldots, m$, we define the indicator random variable

$$\begin{aligned} Y_i &= I\{\text{ clause } i \text{ is satisfied }\} \\ &= 1 - 1/8 \end{aligned}$$

Let $Y$ be the number of satisfied clauses overall, so that $Y = Y_1 + Y_2 + \cdots + Y_m$.

# Randomization

**Proof.**

Then we have

$$E[Y] = E\left[\sum_{i=1}^{m} Y_i\right]$$
$$= \sum_{i=1}^{m} E[Y_i] = \sum_{i=1}^{m} 7/8 = 7m/8$$

Hence the approximation ratio is at most $m/(7m/8) = 8/7$. □

# Randomization

**Proof.**

Then we have

$$E[Y] = E\left[\sum_{i=1}^{m} Y_i\right]$$
$$= \sum_{i=1}^{m} E[Y_i] = \sum_{i=1}^{m} 7/8 = 7m/8$$

Hence the approximation ratio is at most $m/(7m/8) = 8/7$. □

# Linear programming

## Definition

In the ***minimum-weight vertex-cover problem***, we are given an undirected graph $G = (V, E)$ in which each vertex $v \in V$ has an associated positive weight $w(v)$.
For any vertex cover $V' \subseteq V$, we define the weight of the vertex cover $w(V') = \sum_{v \in V'} w(v)$.
The goal is to find a vertex cover of minimum weight.

# Linear programming

## 0-1 integer program

The following $0-1$ integer program is used to find a minimum-weight vertex cover:

**minimize** $$\sum_{v \in V} w(v)x(v)$$

**subject to**

$$\begin{aligned} x(u) + x(v) &\geq 1 && \text{for each } (u,v) \in E \\ x(v) &\in \{0,1\} && \text{for each } v \in V. \end{aligned}$$

# Linear programming

## linear-programming relaxation

Suppose, we remove the constraint that $x(v) \in \{0, 1\}$ and replace it by $0 \leq x(v) \leq 1$.

**minimize** $\sum_{v \in V} w(v)x(v)$

**subject to**

$$\begin{aligned} x(u) + x(v) &\geq 1 && \text{for each } (u, v) \in E \\ x(v) &\leq 1 && \text{for each } v \in V \\ x(v) &\geq 0 && \text{for each } v \in V \end{aligned}$$

# Linear programming

APPROX-MIN-WEIGHT-VC$(G, w)$

```
C ← ∅
compute x̄, an optimal solution
    to the linear program.
for each v ∈ V
    do if x̄(v) ≥ 1/2
        then C ← C ∪ {v}
return C
```

# Linear programming

**Theorem 35.7**

Algorithm APPROX-MIN-WEIGHT-VC is a polynomial-time 2-approximation algorithm for the minimum-weight vertex-cover problem.

# Linear programming

**Proof.**

Let $C^*$ be an optimal solution to the minimum-weight vertex-cover problem, and let $z^*$ be the value of an optimal solution to the linear program.
Since an optimal vertex cover is a feasible solution to the linear program, we have $z^* \leq w(C^*)$.

# Linear programming

## Proof.

$$
\begin{aligned}
z^* &= \sum_{v \in V} w(v)\overline{x}(v) \\
&\geq \sum_{v \in V : \overline{x}(v) \geq 1/2} w(v)\overline{x}(v)
\end{aligned}
$$

□

# Linear programming

**Proof.**

$$z^* \geq \sum_{v \in V : \overline{x}(v) \geq 1/2} w(v) \cdot \frac{1}{2}$$

$$= \sum_{v \in C} w(v) \cdot \frac{1}{2} = \frac{1}{2} w(C)$$

Thus, $w(C) \leq 2z^* \leq 2w(C^*)$. □

# An exponential-time exact algorithm

```
EXACT-SUBSET-SUM(S, t)
n ← |S|
L_0 ← ⟨0⟩
for each i ← 1 to n
      do L_i ← MERGE-LISTS(L_{i−1}, L_{i−1} + x_i)
         remove from L_i every element that
            is greater than t
return the largest element in L_n
```

# An exponential-time exact algorithm

EXACT-SUBSET-SUM$(S, t)$

$n \leftarrow |S|$
$L_0 \leftarrow \langle 0 \rangle$
**for** each $i \leftarrow 1$ **to** $n$
    **do** $L_i \leftarrow$ MERGE-LISTS$(L_{i-1}, L_{i-1} + x_i)$
        remove from $L_i$ every element that is greater than $t$
**return** the largest element in $L_n$

# An exponential-time exact algorithm

## Example

If $S = \{1, 4, 5\}$, then

$$\begin{aligned} P_1 &= \{0, 1\}, \\ P_2 &= \{0, 1, 4, 5\}, \\ P_3 &= \{0, 1, 4, 5, 6, 9, 10\}, \end{aligned}$$

# A fully polynomial-time approximation scheme

## A trimming scheme

- We use a trimming parameter $\delta$ such that $0 < \delta < 1$.
- To ***trim*** a list $L$ by $\delta$ means to remove as many elements from $L$ as possible,

# A fully polynomial-time approximation scheme

## A trimming scheme

- For every element $y$ that was removed from $L$, there is an element $z$ still in $L'$ that approximates $y$, that is,

$$\frac{y}{1+\delta} \leq z \leq y.$$

- $\delta = 0.1, L = \langle 10, 11, 12, 15, 20, 21, 22, 23, 24, 29 \rangle \Rightarrow$, $L' = \langle 10, 12, 15, 20, 23, 29 \rangle$.

# A fully polynomial-time approximation scheme

$\textsc{Trim}(L, \delta)$

```
m ← |L|
L′ ← ⟨y_1⟩
last ← y_1
for i ← 2 to m
        do if y_i > last · (1 + δ)
                then append y_i onto the end of L′
                        last ← y_i
return L′
```

# A fully polynomial-time approximation scheme

```
APPROX-SUBSET-SUM(S, t, ε)
n ← |S|
L_0 ← ⟨0⟩
for i ← 1 to n
        do L_i ← MERGE-LISTS(L_{i−1}, L_{i−1} + x_i)
           L_i ← TRIM(L_i, ε/2n)
           remove form L_i every element that
              is greater than t
let z* be the largest value in L_n
return z*
```

# A fully polynomial-time approximation scheme

## Example

Suppose we have the instance $S = \langle 104, 102, 201, 101 \rangle$ with $t = 308$ and $\epsilon = 0.40$. The trimming parameter $\delta$ is $\epsilon/8 = 0.05$.

# A fully polynomial-time approximation scheme

## Example

APPROX-SUBSET-SUM computes the following values on the indicated lines:

$$\begin{aligned} \text{line } 2: \quad L_0 &= \langle 0 \rangle \\ \\ \text{line } 4: \quad L_1 &= \langle 0, 104 \rangle \\ \text{line } 5: \quad L_1 &= \langle 0, 104 \rangle \\ \text{line } 6: \quad L_1 &= \langle 0, 104 \rangle \end{aligned}$$

# A fully polynomial-time approximation scheme

## Example

$$\begin{array}{lcl} \text{line } 4: & L_2 & = & \langle 0, 102, 104, 206 \rangle \\ \text{line } 5: & L_2 & = & \langle 0, 102, 206 \rangle \\ \text{line } 6: & L_2 & = & \langle 0, 102, 206 \rangle \end{array}$$

$$\begin{array}{lcl} \text{line } 4: & L_3 & = & \langle 0, 102, 201, 206, 303, 407 \rangle \\ \text{line } 5: & L_3 & = & \langle 0, 102, 201, 303, 407 \rangle \\ \text{line } 6: & L_3 & = & \langle 0, 102, 201, 303 \rangle \end{array}$$

# A fully polynomial-time approximation scheme

## Example

$$\begin{aligned}
\text{line } 4: \quad L_4 &= \langle 0, 101, 102, 201, 203, 302, 303, 404 \rangle \\
\text{line } 5: \quad L_4 &= \langle 0, 101, 201, 302, 404 \rangle \\
\text{line } 6: \quad L_4 &= \langle 0, 101, 201, 302 \rangle
\end{aligned}$$

# A fully polynomial-time approximation scheme

**Theorem 35.8**

APPROX-SUBSET-SUM is a fully polynomial-time approximation scheme for the subset-sum problem.

# A fully polynomial-time approximation scheme

**Proof.**

Let $y^* \in P_n$ denote an optimal solution to the subset-sum problem. By induction on $i$, it can be shown that for every element $y$ in $P_i$ that is at most $t$, there is a $z \in L_i$ such that

$$\frac{y}{(1+\epsilon/2n)^i} \leq z \leq y.$$

# A fully polynomial-time approximation scheme

**Proof.**

Thus

$$\begin{aligned}
\frac{y^*}{z^*} &\le \left(1+\frac{\epsilon}{2n}\right)^n \\
&\le e^{\epsilon/2} \qquad \left(\lim_{n\to\infty}(1+x/n)^n = e^x\right) \\
&\le 1+\epsilon/2+(\epsilon/2)^2 \quad (1+x<e^x<1+x+x^2) \\
&\le 1+\epsilon
\end{aligned}$$

# A fully polynomial-time approximation scheme

**Proof.**

After trimming, successive elements $z$ and $z'$ of $L_i$ must have the relationship $z'/z > 1 + \epsilon/2n$. The number of elements in each list $L_i$ is at most

# A fully polynomial-time approximation scheme

**Proof.**

$$\begin{aligned}\log_{1+\epsilon/2n} t + 2 &= \frac{\ln t}{\ln(1+\epsilon/2n)} + 2 \\ &\le \frac{2n(1+\epsilon/2n)\ln t}{\epsilon} + 2 \qquad \left(\frac{x}{1+x} \le \ln(1+x)\right) \\ &\le \frac{4n\ln t}{\epsilon} + 2\end{aligned}$$

□